

Requirements / Specification Document

TITLE: Vector Network Analyzer Extender Heads

Requesting Laboratory / Division / Group:

Communications Technology Laboratory
RF Technology Division
Guided Wave Electromagnetics Group (672.01)

General Statement of Need:

NIST requires two (2) Vector Network Analyzer (VNA) extender heads for WR6.5 with attenuators, and one (1) WR6.5 calibration kit. All equipment must include a 1-year warranty with maintenance. The objective is to enable measurements of integrated circuits that include millimeter wave filters and calibration structures for 110 GHz to 170 GHz. This requirement supports NIST's Innovation in Measurement Science Program.

Item Specifications and Quantities:

Item 1:	WR6.5 Tx/Rx VNA Extender Head Module	
Quantity:	2 EACH	
Specifications:	1.1	110 GHz to 170 GHz bandwidth
	1.2	120 dB dynamic range
	1.3	0.25 dB magnitude stability
	1.4	4 deg phase stability
	1.5	13 dBm maximum test port output power
	1.6	30 dB directivity
	1.7	Integrated Micrometer Driven Output Attenuator
Item 2:	WR6.5 Calibration Kit	
Quantity:	1 EACH	
Specifications:	2.1	TRL Kit (2 short circuit, 2 matched load, 1 straight waveguide, 1 quarter wave shim)

[List any additional item(s) or service requirements using the same format.]

Vendor Requirements

1. Possess at time of award/contracting an Export Control Classification Number (ECCN).

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2. Possess at time of award/contracting a Unique Entity Identification Number (UEI) which registration is listed as “Active” on the GSA System for Award Management (SAM.gov).

[Provide input for each section Below or Mark Not Applicable (N/A)]

Section	Requirements/Specifications
Planning Considerations ☒ N/A	☐ List any visits required to allow potential vendors to see site specific constraints. [Explain Requirements.] ☐ US citizen required on-site (foreign nationals require pre-registration 30 days prior to visit.) ☐ Vendor Representative On-Site more than three (3) days. ☐ Other Considerations: [List other considerations to ensure an accurate quote.]
Shipping & Delivery ☐ N/A	Shipping Address: ☒ NIST-Boulder ATTN: Nate Orloff Mail Stop: 672.01 325 Broadway Boulder, CO 80305 ☐ NIST-Gaithersburg ATTN: Mail Stop: 100 Bureau Dr Gaithersburg MD 20899 ☐ [Alternate Shipping Address] Shipping Criteria: ☐ Partial Delivery Acceptable ☐ [Enter Shipping and Delivery information by item.] ☐ Direct Delivery to OU Building Required [Explain Need for Requirement] Building /Room Number: [Enter Building/Room Number] ☐ Building has Loading Dock Containerization Preference: [Enter Container Type] Other Requirements: [List Any Other Requirement]

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Delivery Date Criteria:

- ☐ Delivery Shall Be Completed No Later Than:
Click or tap to enter a date.
- ☐ Delivery Shall Be Made No Earlier Than:
Click or tap to enter a date.
- ☐ Scheduled Deliveries:

Item Number	Due Date

- ☐ Other:
[List Other Shipping, Delivery or Special Requirements.]

Electronic Media

☒ N/A

[Enter Electronic Media Requirements by Item.]

- ☐ Software
- ☐ Electronic Manuals
- ☐ License Key
- ☐ Other:

[List Any Other Electronic Media Requirements]

- ☐ US Government Email Address for Software Delivery:
[Enter Address]

Installation

☒ N/A

[Enter Installation Requirements by Item]

- ☐ Vendor Technician on Site
- ☐ Rigging
- ☐ Uncrating / Unpackaging
- ☐ Removal of Packaging Material
- ☐ Equipment Set Up
- ☐ Start-Up Services
- ☐ Turnkey Installation
- ☐ Other:

[Enter Details]

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NOTE: TPOC shall be the coordination POC.

Facility / Utility Considerations

☒ N/A

[List Site Specific Utility Considerations by Item]

☐ OFPM Work Order Numbers

[List Work Order Numbers]

☐ Power Available

Voltage: [Enter voltage] (V)

Maximum Current: [Enter max current draw] (A)

Phase: [Choose Phase]

Frequency: [Typically 60Hz] (Hz)

NEMA Plug Type: [Enter Plug Type] (i.e.: NEMA L5-20P)

Environment: [Choose Environment]

Other Power Considerations:

[List Any Other Power Considerations.]

☐ Utility Connections:

[List Relevant Available Utilities, Pressure(s) and Flowrate(s).]

☐ Other Considerations:

[List Other Considerations to Ensure an Accurate Quote.]

Inspection and Acceptance

☐ N/A

NIST will perform measurements on site. NIST will measure the S-parameters of the calibration kit with the extender heads.

☒ On-Site

☐ Acceptance Not Expected to Exceed 7 Business Days

☒ Acceptance Expected to Exceed 7 Business Days:

Acceptance expected to require 14 business days post-delivery.

☒ Inspection and Acceptance Plan:

NIST will measure the S-parameters of the calkit with the extender heads.

☐ Other:

[Enter Other Requirements]

Training

☒ N/A

Training Requirements (Who, When, What, Where)

☐ Number of Trainees: [Enter Number of Trainees]

☐ On-Site

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☐ Off-Site Location: [Enter Location(s)]
Is Trainee Travel Required: [Choose Answer]

Scope of Training

☐ Operation ☐ Maintenance ☐ Troubleshooting
☐ Safety ☐ Other

[Enter Other Scope of Training Requirements]

☐ Training Materials to be Provided by Government:

[Describe Materials]

☐ Training Materials to be Provided by Vendor:

[Describe Materials]

☐ Other

[Enter Other Training Requirements]

NOTE: SME shall be coordination POC.

Warranty

☐ N/A

Base Manufacturer Warranty is acceptable

Warranty Length: 1 Year

What is included?

☒ Labor ☒ Parts ☐ Travel

☒ Cost and Liability for Returns

☐ Other

[Enter Other Requirements]

☐ 1 Option Period

☐ 2 Option Periods

☐ 3 Option Periods

☐ 4 Option Periods (Maximum)

Calibration

☒ N/A

[Describe Calibration Requirements Per Item]

Calibration Plan Length: [i.e. 3 - Years]

What is Included?

☐ Labor ☐ Spare Parts ☐ Consumables

☐ Travel ☐ Cost and Liability for Shipment

☐ Software Updates ☐ Manuals

☐ Other

[Enter Other Requirements]

☐ Requirements:

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[Enter Requirements] (i.e.: On-Site Within 72 hours, etc.)

☐ 1 Option Period

Length: [i.e. 1 – Year / 6 - Months]

☐ 2 Option Periods

☐ 3 Option Periods

☐ 4 Option Periods (Maximum)

Maintenance

☒ N/A

[Describe Maintenance Requirements Per Item]

Maintenance Plan Length: [i.e. 3 - Years]

What is Included?

☐ Labor ☐ Spare Parts ☐ Consumables

☐ Travel ☐ Cost and Liability for Shipment

☐ Software Updates ☐ Manuals

☐ Other

[Enter Other Requirements]

☐ Requirements:

[Enter Requirements] (i.e.: On-Site Within 72 hours, etc.)

☐ 1 Option Period

Length: [i.e. 1 – Year / 6 - Months]

☐ 2 Option Periods

☐ 3 Option Periods

☐ 4 Option Periods (Maximum)

Government
Furnished
Material /
Equipment

☒ N/A

List Any Government Furnished Material / Equipment Per Item]

☐ Samples for Testing

☐ Equipment which will leave Government Site

Description: [Enter Description]

Serial Number: [Enter Serial Number]

NIST Property Number: [Enter NIST Property Number]

☐ Other: [Enter Other Details]

Travel

☒ N/A

[Describe any Travel Requirements Per Item]

☐ Explain:

[Enter Justification / Requirement]

☐ Other:

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[Enter Other Requirements]

NOTE TO VENDOR: To the maximum extent possible, include travel requirement estimates as part of pertinent sections above (Installation, Maintenance, Repair, etc.). Otherwise, Travel will be a separate line item, reimbursable per the Federal Travel Regulation.